

Static vs Contextual Embeddings

for Lexical Ambiguity: A Word-in-Context Evaluation

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01 The concept

Lexical ambiguity means a word has several meanings. **Context**, the surrounding words, helps identify the intended meaning.

Homonymy: unrelated meanings [4]. A river **bank** and a savings **bank** refer to different things. **Polysemy: related meanings** [4], such as “newspaper” for the printed object or its publisher.

Word embeddings are vectors: lists of numbers representing words. They are learned from language data, with similar word uses often producing similar vectors.

03 Related approaches

Word sense disambiguation selects a dictionary meaning. Knowledge-based methods use definitions and word relations. Supervised methods learn from sense-labelled examples [5].

Word-in-Context (WiC) instead asks whether two uses share a meaning [1]. We test pretrained vectors on this task.

04 Hypothesis

BERT will distinguish word meanings more accurately than static GloVe. Sentence context should help separate meanings that share a fixed word vector.

02 Static vs contextual

Static embeddings: one vector per word. GloVe learns from word co-occurrence counts [2]. Its vector for “bank” stays the same in river and money sentences. Both meanings share one vector.

Contextual embeddings: one vector per use. BERT uses attention to combine information from surrounding words [3]. It builds a context-dependent vector for each occurrence of “bank.”

Both learn from context. BERT also updates the target representation for each new sentence.

05 Methodology

1. Represent each use. Compare GloVe target only, GloVe nearby-word average and BERT target in context.

2. Compare vectors. Cosine similarity measures how closely their directions align. Above a cutoff, predict “same meaning.”

3. Evaluate. Select settings and cutoffs on 5,428 training pairs. Score 638 separate English validation pairs, half same meaning and half different.

Settings: GloVe 6B, 300 dimensions, up to two words on each side. BERT-base-uncased, mean of target subwords in the last four layers. No BERT fine-tuning.

06 Results

BERT answered 74 more pairs correctly.

GloVe: word only

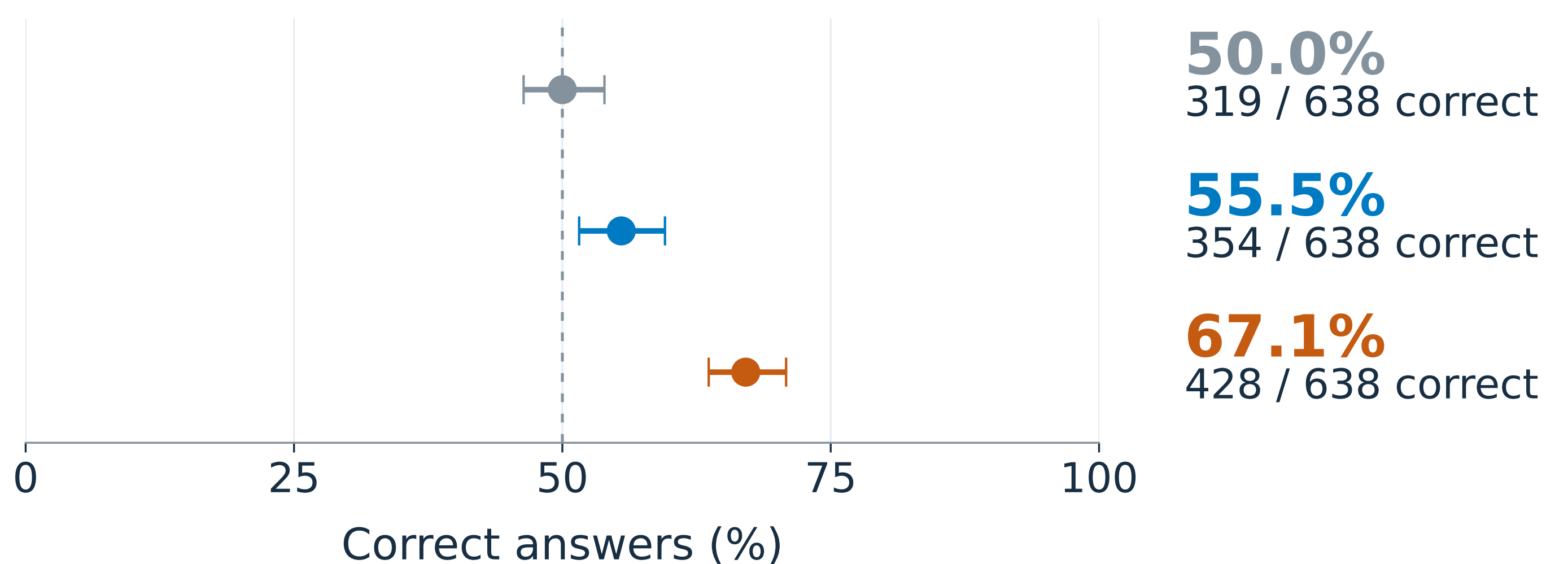
One fixed vector for the target word

GloVe: nearby words

Average of nearby word vectors

BERT: word in its sentence

Target vector changes with context



Dots show accuracy and short lines show 95% bootstrap intervals. The dashed line marks the 50% chance level.

BERT’s advantage: 11.6 percentage points (95% paired interval: 6.6 to 16.9). Uncertainty estimated from 1,000 resamples of the validation pairs.

07 Conclusion

Contextual word representations help distinguish meanings on WiC. BERT’s 67.1% accuracy supports our hypothesis. Nearby-word GloVe reaches 55.5%. BERT still makes 210 errors, so context helps without resolving every ambiguous use.

References [1] Pilehvar & Camacho-Collados (2019). *WiC: the Word-in-Context Dataset*. NAACL.

[2] Pennington, Socher & Manning (2014). *GloVe: Global Vectors for Word Representation*. EMNLP.

[3] Devlin et al. (2019). *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*. NAACL.

[4] Haber & Poesio (2021). *Patterns of Polysemy and Homonymy in Contextualised Language Models*. Findings of EMNLP.

[5] Loureiro et al. (2021). *Analysis and Evaluation of Language Models for Word Sense Disambiguation*. Computational Linguistics.

Methods, additional analyses and reproducibility details: accompanying appendix.